

[ISOM2010] INTRODUCTION TO INFORMATION SYSTEMS

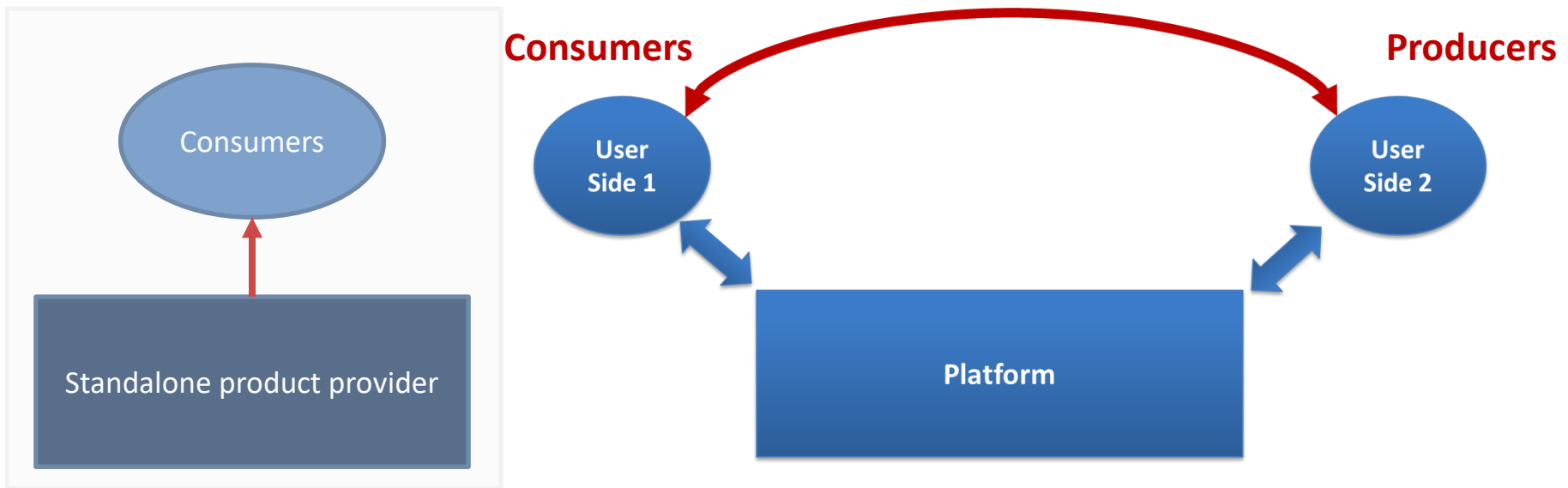
DIGITAL PLATFORMS: NETWORK EFFECTS & PLATFORM ARCHITECTURE

S9

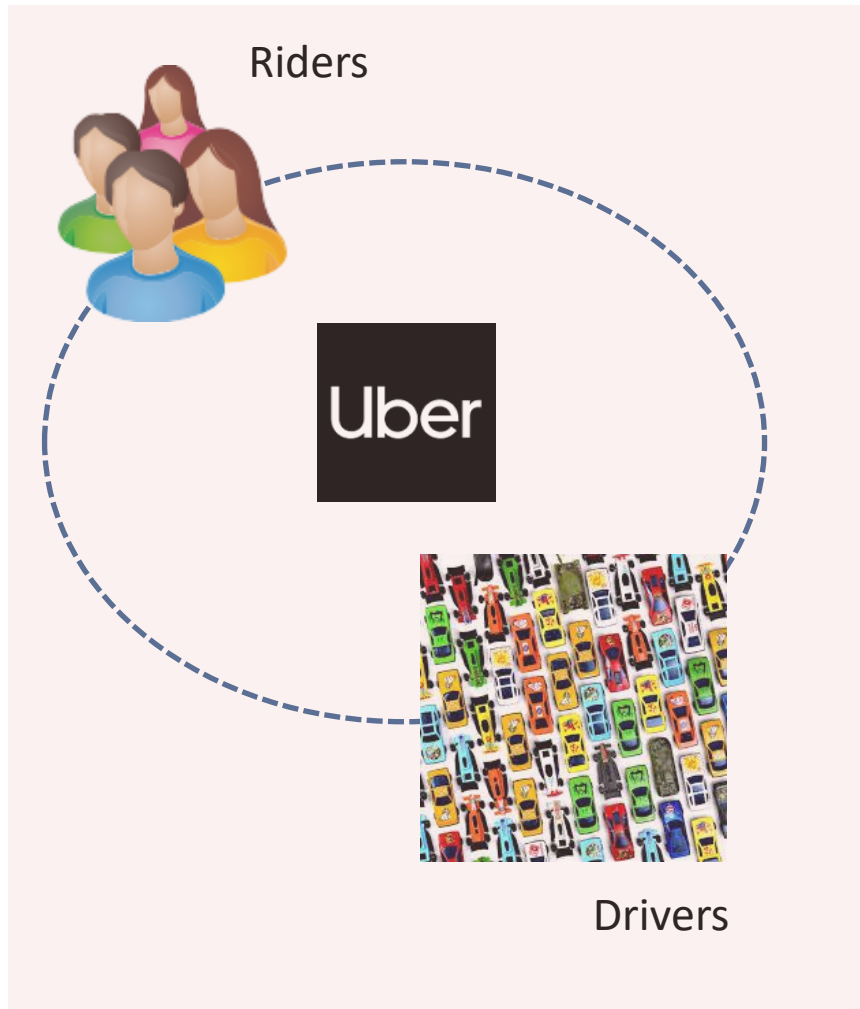
PROF. KIM, YONGSUK

More on the concept of a platform

- A business based on enabling **value-creating interactions** between **external (3rd party) producers** and **consumers**.
- A platform provides an open, participatory infrastructure for these interactions and sets governance conditions for them.
- A platform can be hardware, software, or even a shopping mall.

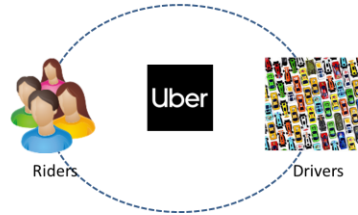


Transaction Platform vs. Innovation Platform



Games are **complementary goods** for a game console;
Gamers (consumers) value them *together*

Transaction Platform vs. Innovation Platform



	Transaction Platform	Innovation Platform
Definition	The platform serves as an intermediary for direct exchange or transactions	The platform serves as a technological foundation upon which other firms develop complementary innovations
Examples	Twitter, Instagram, Uber, LinkedIn, and FoodPanda	Nintendo, Windows, and Apple iOS
Network Effects	Come from the increasing number of participants, functions, content or services	Come from the increasing number or utility of the <u>complements</u> *
Value Creation	By facilitating the buying or selling of goods or other interaction between users	By facilitating the development of new complementary products and services

* Complements are innovations that add functionality or access to assets that make the platform increasingly useful.

Other transaction platform examples?

Other innovation platform examples?

Standalone products vs. platforms



Based on technical quality and features (stand-alone value)



Based on **network effects** AND/OR technical quality and features

Stand-alone value vs. Network effect value

Stand-alone Value

The value of technological utility such as functionalities, simplicity, convenience, design, ease of use, risk, etc.



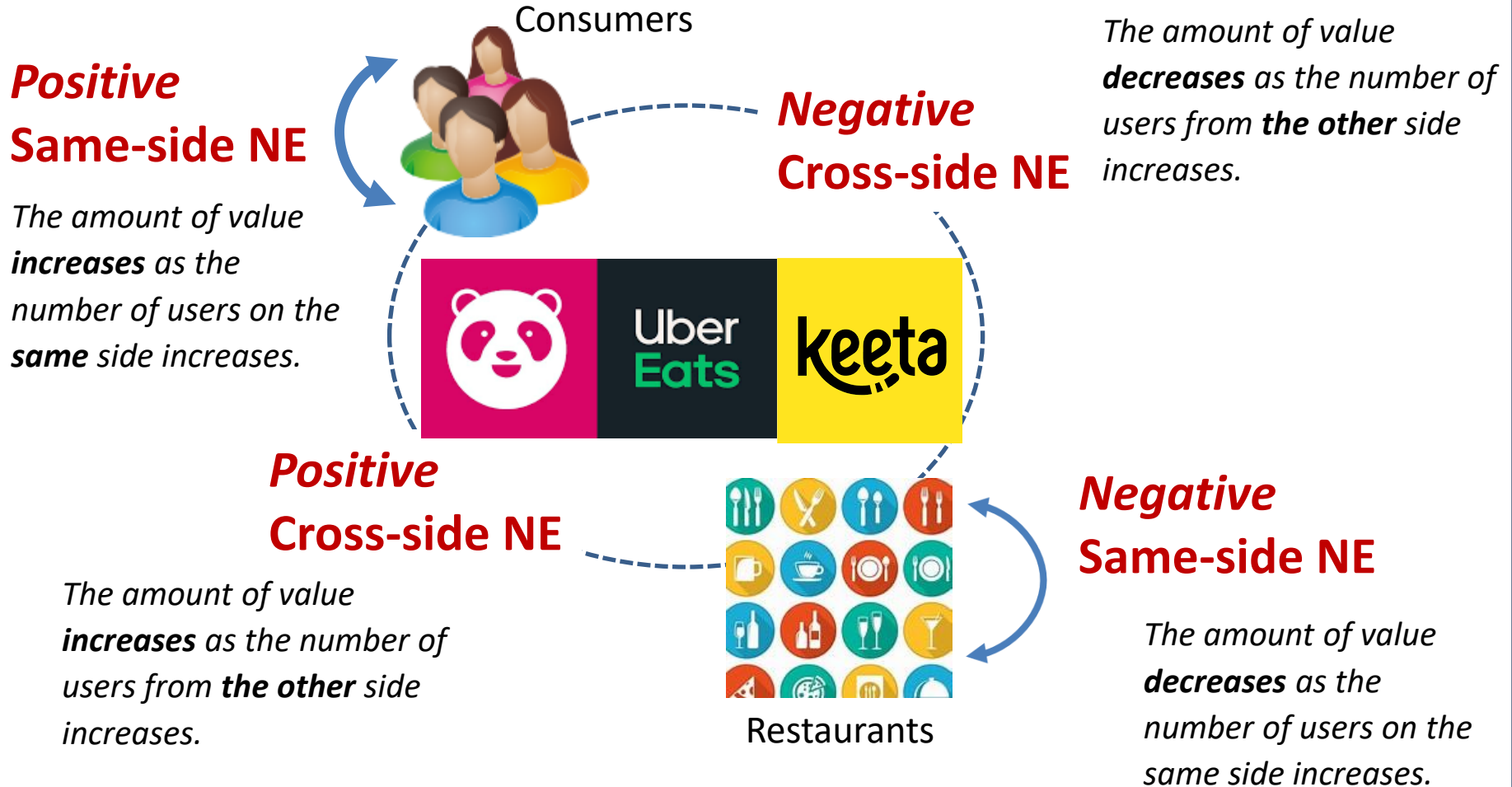
Network Effect Value

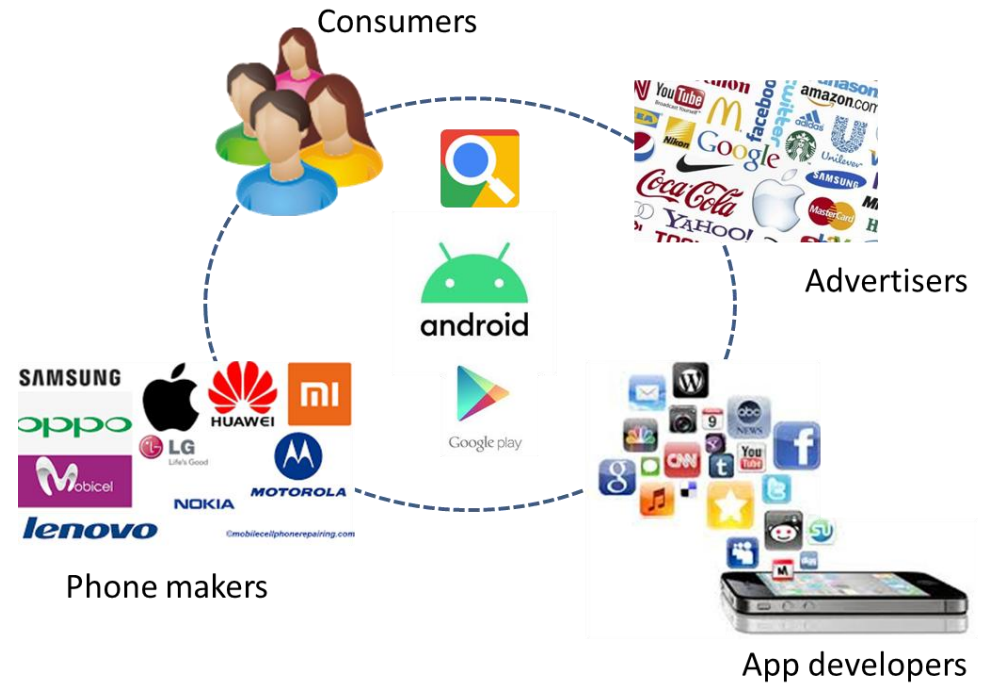
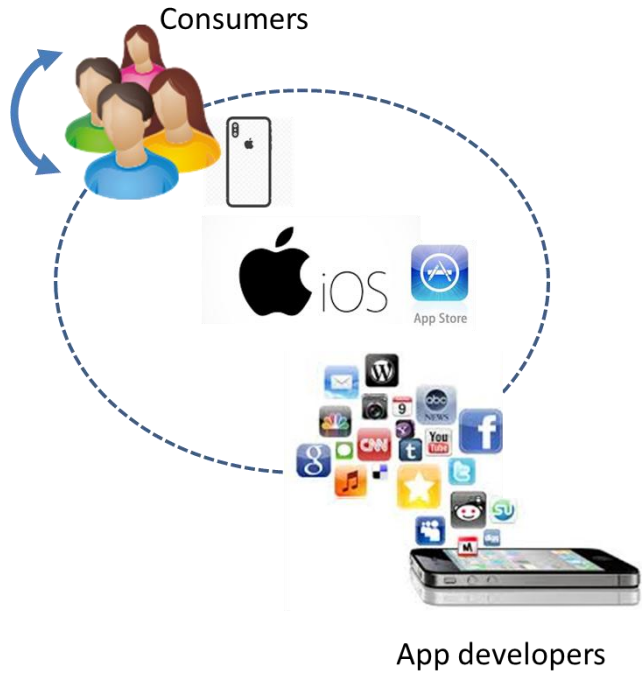
Network effect exists when the value of a “product” increases with the number of other users of the “product”



Let's go deeper into Network Effects

Cross-side and Same-side network effects (NE)

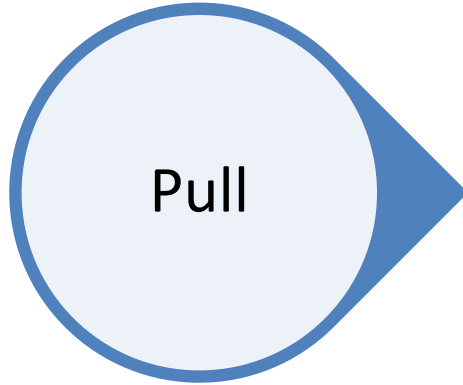




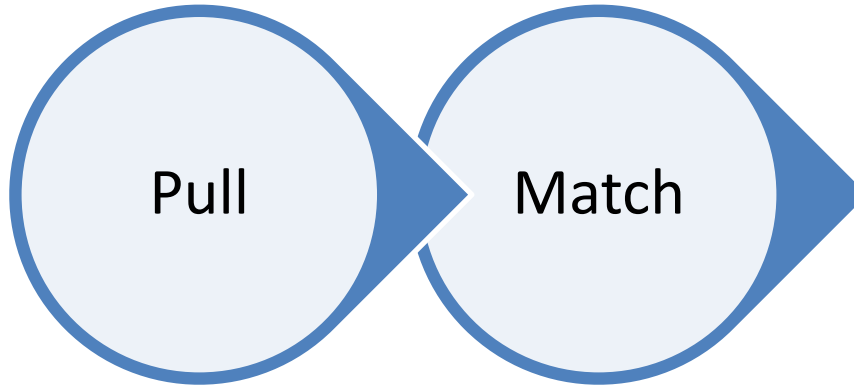
Food ordering platforms



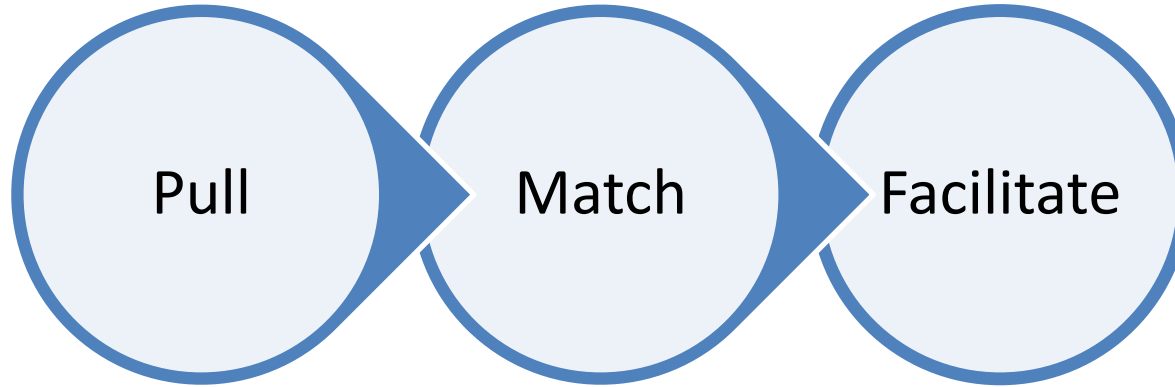
Platform Architecture



Platform Architecture



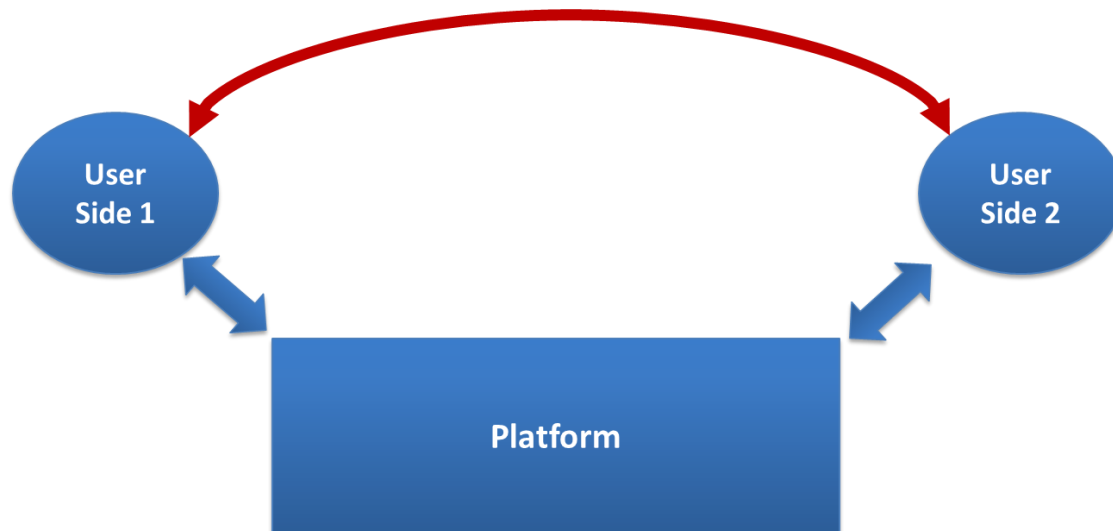
Platform Architecture



The Core Interaction

- The single **most important** form of activity taking place on a platform
 - The exchange of value that attracts MOST users to the platform in the first place

Participants + Value Unit + Filter → Core Interaction



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Participants + **Value Unit** + **Filter** → **Core Interaction**

- Producer
- Consumer

- Created by the producer
- To be exchanged
- Why people interact

- Algorithm-based
- Delivery of the value unit
- Users receive only relevant, valuable value units

- Defines why the platform exists

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Platform	Value Unit
Airbnb	
Twitter	
YouTube	
Kickstarter	
Android	
OpenRice	

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Functions: HOW

PULL

We need to bring *both* sides on board.

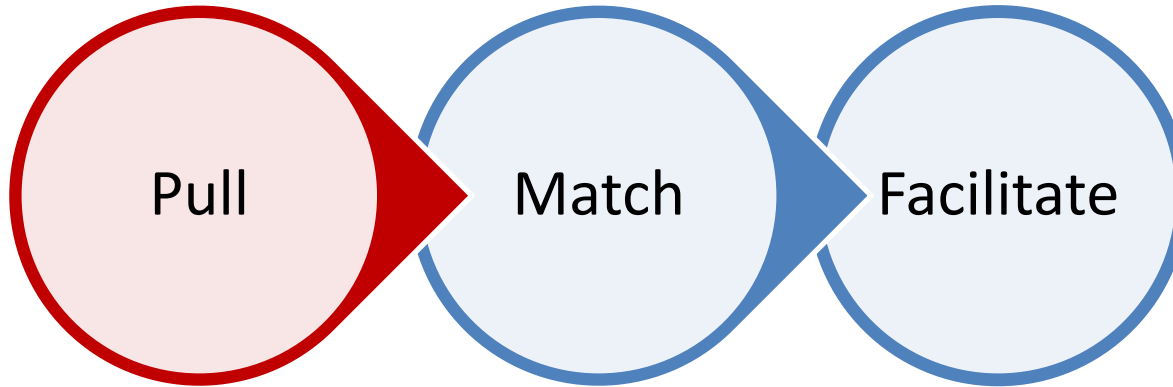
MATCH

We need *data* that helps facilitate the match.

FACILITATE

We need to give *tools* to help them interact with one another and *rules* to encourage valuable exchanges (while discouraging others)

(1) Pull



- Producer
- Consumer
- Value unit

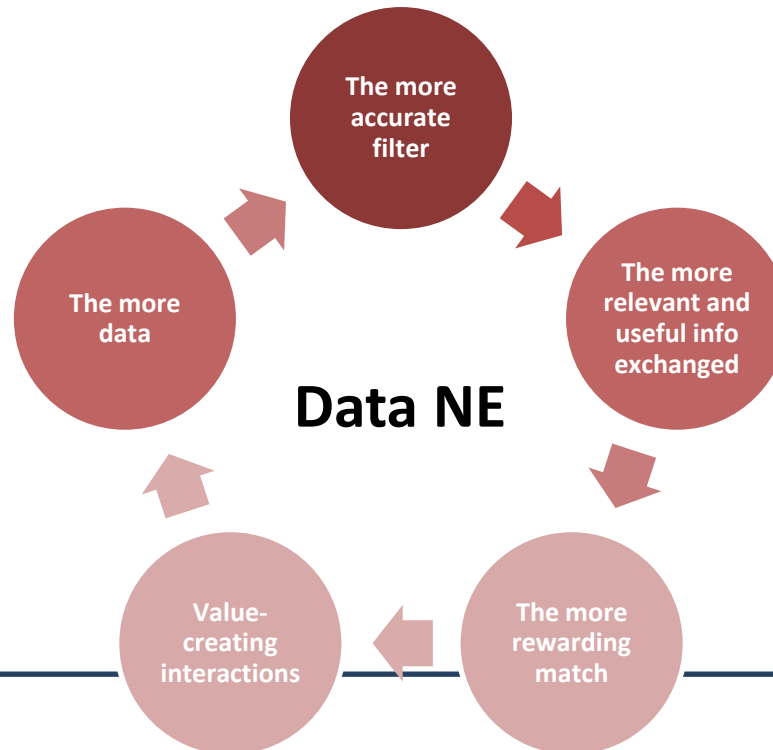
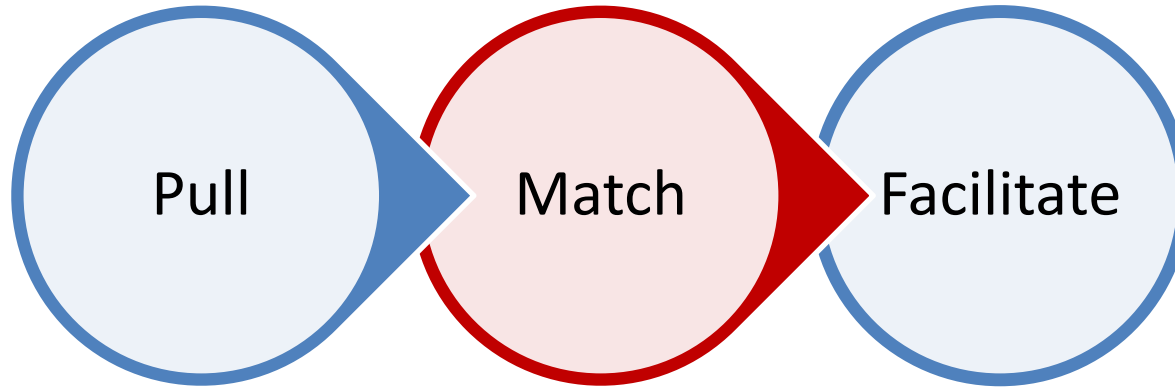
(New) user acquisition

- Identification: B2C vs. C2C
- Chicken-and-egg (CE) problem

(Existing) user retention

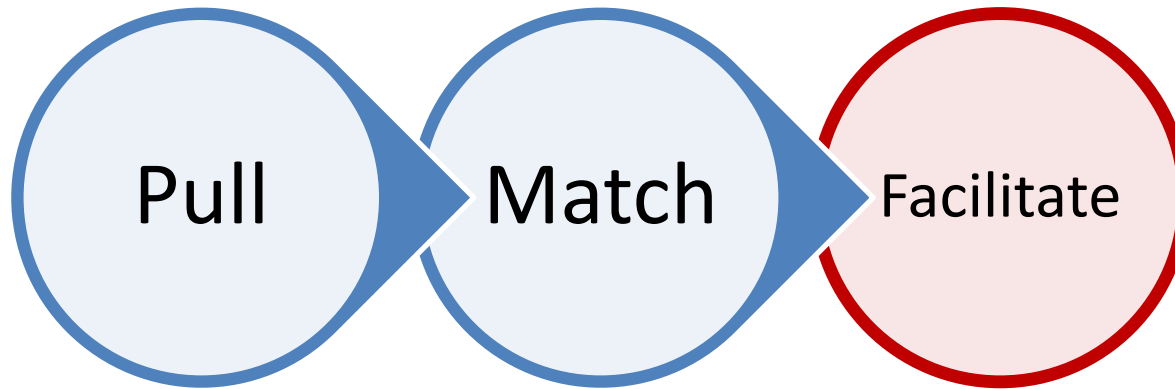
- Value-creating interactions
- Disintermediation
- Multi-homing

(2) Match



- Data issues (Facebook vs. Tinder)
- Matching issues
- Recommendations (prediction)

(3) Facilitate



Facilitation
Tools/Services

Design *tools/services* that make it as easy and safe as possible for producers to create and for consumers to receive valuable goods/services via the platform

Platform
Governance

A set of rules and tools to increase the quality of participants and interactions to build users' trust

Platform Architecture

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